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Induced morphological variation in *Vicia faba* L.

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Morphological observations, and to some extent genetic analyses, are presented for some groups of induced morphological mutations in *Vicia faba* L. They comprised unifoliate (9), flower colour (36) and seed-coat colour (57) mutants; dwarfs (5); types with altered pollen shape (3); a late maturing form (1); red-stemmed seedling (1); and forms with changes in the colour and shape of the hilum (4). Furthermore, an early mutant, one type with a terminal inflorescence, and one with increased protein content are discussed mainly from a plant breeding point of view. When compared with the parental variety, most of the mutants are recessive. Black and violet seed-coat colour are dominant to the buff colour of the parental variety, and red-stemmed seedling character is dominant to the normal green colour.

Chromosome mutations, in contrast to genome mutations, usually involve changes in the chromosomes that do not affect the chromosome number. Chromosome mutations can be divided into two categories: gene mutations and structural rearrangements. Of the different types of structural rearrangements, translocations and inversions were the only types that were observed in the present investigations of induced mutations in *Vicia faba* (see SJÖDIN 1971 a and b). In the present paper, some groups of morphological mutants are discussed both from the genetic aspects and with respect to their possible exploitation in plant breeding.

The nature of the mutational events that have produced these mutants is not known. In lower organisms, mutations at the DNA level have been described (e.g., HESLOT 1965), but in higher organisms the resolving power for genetic analyses is low, and the information available has been mainly gleaned from investigations in *Zea mays*. However, the morphological mutations presented here represent distinct characters distinguishing them from the parental variety Primus. The segregation of these mutations follow a monogenic Mendelian ratio of 3: 1, even though it is not uncommon with a deficit of recessive offspring. No cytological irregularities have been observed in any of the mutants. Although minor deficiencies and duplications cannot be excluded, the mutations have been classified as "gene mutations", which is also a useful term for practical purposes.

In the course of this investigation, mutagenic treatments have been carried out every year, and as a result, there has been a continuous addition of new mutants. Because several years are required to investigate new mutant accessions more thoroughly, the present paper, naturally, only reflects the progress that has been made up to the present time. The main purpose of this presentation is, besides reporting the presently available genetic data, to inform the reader of the available *Vicia faba* mutants in the Svalöv collection.

Spontaneous mutants exhibiting the same characters as the induced mutants have been collected and analysed during the course of this investigation. The data from the spontaneous types are included in the presentation of the respective group of mutants.

Materials and methods

The standard variety that was used in the mutagenic treatments was Svalöf Primus, bred at the Swedish Seed Association. The mutagens em-

ployed were X-rays, neutrons, gamma-rays, ethyl methane sulphonate (EMS), methyl methane sulphonate (MMS), ethylene imine (EI), and EMS + Cu^{++}.

The gene symbols used in the investigation mainly conform with the rules recommended by the International Committee on Genetic Symbols and Nomenclature, which were published in 1957. In those cases in which genetic analyses of *Vicia faba* have been published elsewhere, the same gene symbols have been adopted with one exception – the *O* gene proposed by SIRKS (1931).

Pollen fertility was estimated from microscopic analyses of slide preparations stained with lactophenol fuchsin.

Most linkage data involving marker genes and translocations have been obtained from backcrossing the F_1 to the parent carrying the recessive allele. Furthermore, some crossover values based on classification in the F_2 generation are presented in Table 13. These data were estimated by the product method.

The distributions of phenotypic classes have been tested for goodness of fit before the P-values were calculated. The formulas described by PERSSON (1969) have been used. The segregation was deemed good when the Chi-square test for goodness of fit for each of the segregating factors did not attain the 1 per cent level of significance. When χ^2_L (e.g., Chi-square test for linkage) was significant at the 5 per cent level, the crossover value was determined.

Besides the mutations induced in the present investigation, different morphological types have been obtained from various botanical gardens and through the generosity of other workers: Drs. D. A. BOND, D. G. ROWLANDS, J. PICARD and P. BRYSSINE.

Results

1. Unifoliate mutants (*un*)

Unifoliate mutants have been described in many species of Leguminosae, and have been recently reviewed by GOTTSCHALK (1969). As the name implies, a unifoliate mutant is characterized by a transformation of the typical compound leaf into a simple leaf. The simple, or unifoliate, leaf is more or less pinnately veined; however, it is in most instances perceptibly tripli-nerved as a result of the fusion, or connation, of the three leaflets during morphogenesis. The unifoliate leaf displays great variation in shape (oval, orbiculate, obovate, etc.) and lobation. Some of the variable leaf types are illustrated in a previous paper (SJÖDIN 1964, Fig. 2, p. 282).

Table 1. Origin of the investigated unifoliate mutants in *Vicia faba*

Gene-symbol	Origin
un-a^1	Induced, X-irradiation, 8000 r, 1958
un-a^2	Spontaneous, obtained from GOTTSCHALK, 1961
un-a^3	Induced, X-irradiation, 6000 r, 1961
un-bc^1	Spontaneous, found in Bohuslän, Sweden, 1961
un-a^4	Induced, injection with acetone, 10 %, obtained from BRYSSINE, 1963
un-a^5	Spontaneous, isolated in breeding material, 1966
un-a^6	Spontaneous, isolated in breeding material, 1966
un-a^7	Induced, MMS, 0.05 %, 1965
un-a^8	Induced, neutrons, 140 rad, 1966

The present collection of unifoliate types available at Svalöf is presented in Table 1. The unifoliate mutants that were induced at Svalöf and the spontaneous mutant isolated in Bohuslän (*un-bc*^{1}) originated from the commercial variety Primus. The mutant *un-a*^{2} was first reported by SCHEIBE and GOTTSCHALK (1956). The mutants *un-a*^{1}, *un-a*^{3}, and *un-bc*^{1} have been described both morphologically and genetically (SJÖDIN 1964).

It was found that the mutants could be separated into two groups with regard to the expression of the unifoliate character during the vegetative phase. Some of the mutants (obligate), retain the unifoliate character throughout ontogenesis, whereas others (transnormal) exhibit the unifoliate character during the early stages of development, after which the shape of the leaf gradually reverts to the normal form (see Fig. 1). This gradual transition is usually completed after 6 to 8 leaves are present.

Representative of the transnormal type are *un-bc*^{1} and *un-a*^{4}. During the early generations, *un-a*^{3} displayed the unifoliate character throughout the vegetative development; but this mutant has undergone some modification in later

generations, with the late leaves at the top of the plant manifesting normal shape. Morphologically, *un-a*^{4} is distinguished from the other mutants by its greater tillering capacity, smaller leaves, earlier flowering, and reduced number of flowers (mostly one) per inflorescence. However, all of these characters are most likely primarily phenotypic responses to the changes in day length between Rabat, Morocco (the locality of origin) and Svalöf.

All mutants with the obligate leaf type exhibited a somewhat abnormal development during the late ontogenetic stages. The internode lengths are irregular with most of the internodes shorter than in the standard material, i.e., Primus. It is, moreover, rather characteristic for the upper half to upper third of the mutants to be leafless (Fig. 1). The phyllotaxy of the mutants is not rarely irregular, and then deviating from the normal (180°) opposite leaf arrangement (Fig. 1).

Mutant types are often characterized by modification in several other characters as compared with the standard material (see GOTTSCHALK 1967). These differences may be evoked by genes with pleiotropic effects. The unifoliate mutants of *Vicia faba* also deviate from the parental varieties in more than one character: The structure of the flower is variously modified. The floral modifications especially involve the number of parts. Data from measurements of floral deviations are presented in Table 2. For the sake of surveyability, the data from *un-a*^{1}, *un-a*^{3}, and *un-bc*^{1} are included in the table (from SJÖDIN 1964). Corresponding data for *un-a*^{2} have been reported by SCHEIBE and GOTTSCHALK (1956).

The typical flower of *Vicia faba* is comprised of a standard (vexillum or banner), two wings (alae), and a keel (carina), which is formed by the coalescence of the two lower or anterior petals. Owing to difficulties in determining the exact number of stamens at the stage in which the other floral parts were analysed, stamen data is not presented, However, from observations at this and later stages of floral development, no deviations from the normal number of stamens were found.

Un-bc^1 (Table 2) has a floral composition very close to Primus, whereas the other mutants are characterized by a greater variation in the number of floral parts. *Un-a*^{3} displays a greater variation throughout, while the other mutants deviate more extremely with respect to individual floral elements. The number of ovules in the standard material is on an average four. Except in *un-bc*^{1}, the mutants exhibit a sharp reduction in the number of ovules – with about one-half the number present in Primus.



Fig. 1. Unifoliate mutants of *Vicia faba*. At the left, the parental variety; in the middle, a unifoliate mutant of the obligate type; at the right, the transnormal type.

In addition to the variation in the number of floral parts, the flowers of the unifoliate mutants are often distinguishable from the standard material in other characters. Thus, the two petals comprising the keel are very frequently split. When more than one pistil is present, they are connate, at least basally. Rather frequently, the carpel is split along the ventral suture and the ovules are then exposed. Occasionally, the standard and wings are smaller than in the parental material; and in flowers with more than two wings, one or more of the wings may be enclosed by the keel. And finally, in *un-a*^{7}, about 20 per

Table 2. Number of floral parts in per cent in the parental variety (N) and in the unifoliate mutants of *Vicia faba*

Material	Standard			Wings					Keel					
	0	1	2	1	2	3	4	5	1	2	3	4	5	6
N	0.0	100.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0
un-a^1	0.3	99.4	0.3	0.3	98.5	0.9	0.0	0.3	0.3	99.7	0.0	0.0	0.0	0.0
un-a^3	4.0	94.0	2.0	0.0	38.0	34.0	22.0	6.0	5.0	95.0	0.0	0.0	0.0	0.0
un-bc^1	0.6	99.4	0.0	0.0	100.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0
un-a^4	0.0	99.0	1.0	0.0	88.0	11.0	1.0	0.0	2.0	84.0	14.0	0.0	0.0	0.0
un-a^5	0.0	100.0	0.0	0.0	95.0	3.0	2.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0
un-a^6	1.0	99.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0
un-a^7	1.0	99.0	0.0	0.0	98.0	1.0	1.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0
un-a^8	0.0	97.0	3.0	0.0	97.0	3.0	0.0	0.0	1.0	91.0	6.0	1.0	0.0	1.0

Material	Pistil				Ovules							n
	1	2	3	4	1	2	3	4	5	6	x	
N	100.0	0.0	0.0	0.0	0.0	0.0	18.5	71.0	10.0	0.5	3.9	200
un-a^1	93.2	6.2	0.6	0.0	6.5	75.5	16.8	0.6	0.6	0.0	2.2	340
un-a^3	93.0	5.0	2.0	0.0	18.0	61.0	8.0	12.0	1.0	0.0	2.2	100
un-bc^1	99.4	0.6	0.0	0.0	0.0	0.0	12.3	71.7	15.4	0.6	4.0	163
un-a^4	67.0	29.0	4.0	0.0	20.0	55.0	18.0	7.0	0.0	0.0	2.1	100
un-a^5	62.0	36.0	1.0	1.0	11.0	64.0	21.0	4.0	0.0	0.0	2.2	100
un-a^6	98.0	2.0	0.0	0.0	6.0	86.0	8.0	0.0	0.0	0.0	2.0	100
un-a^7	100.0	0.0	0.0	0.0	5.0	60.0	34.0	1.0	0.0	0.0	2.3	100
un-a^8	99.0	1.0	0.0	0.0	1.0	91.0	7.0	1.0	0.0	0.0	2.1	100

Table 3. Pollen fertility in the parental variety (N) and in the unifoliate mutants of *Vicia faba*

	N	*un-a*^{1}	*un-a*^{2}	*un-a*^{3}	*un-bc*^{1}	*un-a*^{4}	*un-a*^{5}	*un-a*^{6}	*un-a*^{7}	*un-a*^{8}
Pollen fertility in per cent	94.9	84.1	94.4	86.0	94.9	90.3	95.0	91.6	95.0	95.0
Number of plants examined	200	339	200	144	163	98	100	100	100	100

cent of the keels are twisted as a result of one of the keel petals being smaller.

Pollen fertility has been estimated (see Table 3). Some of the mutants exhibited a reduction in pollen fertility by comparison with the standard material. This reduction in pollen fertility may, however, be assumed to influence the seed fertility only to a minor degree. No measurements of seed fertility have been carried out. Because of the marked reduction in the number of ovules per flower, a reduction in seed set occurs, and the most frequent observation is only one seed per pod. Earlier (SJÖDIN 1964), it was demonstrated in reciprocal crosses between *un-a*^{1} and standard material that the pistillate parts are the principal determinants as regards the low seed-setting ability.

GOTTSCHALK (1958) found that *un-a*^{2} was conditioned by a single recessive gene. Single recessive genes were also responsible for the mutant characteristics in *un-a*^{1} and *un-a*^{3} (SJÖDIN 1964). These three genes were verified as multiple alleles

Table 4. Origin of the group of flower-colour mutants in *Vicia faba*

Gene symbol	Phenotype of flowers	Origin
dp-a^1	violet	EI 0.0031%, 1958
dp-a^2	violet	X-irradiation, 1956
dp-a^3	dark brown	X-irradiation, 1956
dp-a^4	dark brown	X-irradiation, 1956
dp-a^5	dark brown	X-irradiation, 1956
dp-a^6	dark brown	X-irradiation, 1956
dp-a^7	pink standard, violet wings	X-irradiation, 6000 r, 1958
dp-a^8	light pink standard, brown wings	X-irradiation, 8000 r, 1957
dp-a^9	pink standard, brown wings	EI 0.0031%, 1959
dp-a^{10}	light pink standard, brown wings	EI 0.0062%, 1959
dp-a^{11}	violet standard, brown wings	EI 0.025%, 1959
dp-a^{12}	violet	EI 0.025%, 1959
dp-a^{13}	dark brown	EI 0.025%, 1959
dp-a^{14}	dark brown	X-irradiation, 7000 r, 1959
dp-a^{15}	violet	X-irradiation, 7000 r, 1959
dp-a^{16}	dark brown	X-irradiation, 7000 r, 1959
dp-a^{17}	dark brown	Spontaneous, from Bohuslän, 1961
dp-a^{18}	violet	Spontaneous, from Bohuslän, 1961
dp-a^{19}	dark brown	Neutrons, 132 rad, 1961
dp-a^{20}	pink	Spontaneous, obtained from BOND (9311/1/1/1), 1963
dp-a^{21}	dark brown	Spontaneous, obtained from BOND (9311/1/2/3), 1963
dp-a^{22}	violet	Spontaneous, obtained from BOND (51/3 × 9311), 1963
dp-a^{23}	scarlet	Spontaneous, obtained from BOND (C5/2/12/1), 1963
dp-a^{24}	dark brown	Spontaneous, obtained from BOND (C8/2/2/1/1/1), 1963
dp-a^{26}	dark brown	Spontaneous, obtained from ROWLANDS (VI 63/8), 1964
dp-a^{27}	dark brown	Spontaneous, obtained from ROWLANDS (CI), 1964
dp-a^{28}	dark brown	Spontaneous, obtained from ROWLANDS (C8), 1964
dp-a^{29}	solid yellow	Spontaneous, obtained from ROWLANDS (IV 63/2), 1964
dp-b^1	yellow wing spots	Spontaneous, obtained from ROWLANDS (VI 6309), 1964
dp-31	greyish brown	Gamma-rays, 9000 r, 1967
sp-a^1	white	Spontaneous, obtained from BOND, 1963
sp-a^2	white	Spontaneous, obtained from ROWLANDS (Ch 170), 1964
sp-b^1	white	Spontaneous, obtained from ROWLANDS (Triple White), 1964
sp-b^2	white	Spontaneous, obtained from ROWLANDS (VI 6302), 1964
sp-b^3	white	Spontaneous, obtained from ROWLANDS (VI 6301), 1964
sp-b^4	white	Spontaneous, obtained from PICARD, 1968

of the same locus, whereas *un-bc*^{1} was believed to be determined by two other loci. *Un-bc*^{1} gave a segregation ratio very close to 15 : 1 when crossed with the standard material. *Un-a*^{4} and *un-a*^{8} have not yet been tested in crosses with the standard material. *Un-*^{5}, *un-a*^{6}, and *un-a*^{7} showed segregation ratios in close agreement with a 3 : 1 Mendelian segregation. Thus, *un-a*^{5} segregated in 81 normal: 25 *un* ($\chi^2 = 0.113$; $0.95 > P > 0.90$); *un-a*^{6}, in 84 normal: 35 *un* ($\chi^2 = 1.235$; $0.30 > P > 0.20$); and *un-a*^{7}, in 34 normal: 9 *un* ($\chi^2 = 0.380$; $0.70 > P > 0.50$). The five genes *un-a*^{4}, *un-a*^{5}, *un-a*^{6}, *un-a*^{7} and *un-a*^{8} displayed the unifoliate character when crossed with *un-a*^{1}, *un-a*^{2} and *un-a*^{3}, whereas all F_1 offspring manifested normal characteristics when the five mutants were crossed with *un-bc*^{1}. These results signify that all the mutants except *un-bc*^{1} are multiple alleles of the same locus.

2. Flower colour mutants (*dp, sp*)

The typical flower of *Vicia faba* is white with two black spots, or blotches, on the wings and with dark brown or violet lines, or striae, on the standard. The lines on the standard vary in intensity as well as in frequency for each of the colours. Up to now, four loci have been found that affect flower colour. They are designated *sp-a*, *sp-b*, *dp-a*, and *dp-b* (Table 4). Two of these, *sp-a* and *sp-b*, are functionally independent basic loci for

development of colour. *Dp-a* is in the homozygous recessive state characterized by solid flower colour, which can range from very dark brown to violet, i.e., the same colours that constitute the lines on the standard in the typical flower. One mutant conferring solid yellow flowers is also an allele within the *dp-a* locus. The fourth locus, *dp-b*, contains only one mutant, which in the homozygous recessive state exhibits yellow spots on the wings as compared with the usual dark brown or blackish ones.

The six white-flowering types, none of which have been collected in the present mutation material (Table 4), are characterized by the total absence of pigment in the petals. Thus, they lack the black spots on the wings and also the corresponding spots on the stipules. Two multiple alleles have been found in the *sp-a* locus and four in the *sp-b* locus (see Table 4). When representatives of these two loci were crossed to each other, the F_1 gave rise to plants with flower colour typical for the standard variety. The F_2 segregated into 75 plants with normal flowers and 61 plants with white flowers. This segregation fits a 9 : 7 ratio closely ($\chi^2 = 0.0672$; $0.80 > P > 0.70$), indicating a complementary interaction of the alleles from these two loci. This segregation ratio verifies the results obtained by ROWLANDS *et al.* (1962) when they crossed *sp-a*^{2} with *sp-b*^{1}. Finally, it is also the same ratio as that which BATESON (1913) obtained already in 1909 when he crossed two white-flowered varieties of *Lathyrus odoratus* L. (sweet pea).

When plants exhibiting the mutant character from each of these two loci were crossed to the standard variety, Primus, the F_1 displayed normal flowers, indicating that the mutants were recessive to normal flowers. The F_2 generation from the cross Primus × *sp-a* also indicated a 3 : 1 segregation (26 : 6; $\chi^2 = 0.667$; $0.50 > P > 0.30$) whereas the F_2 generation from the cross Primus × *sp-b* segregated in 82 Primus-like plants (normal) and 49 plants with white flowers ($\chi^2 = 10.8$; $P = 0.001$). The reason for this excess of white-flowered plants is not known.

In locus *dp-a*, 28 mutants, forming a series of multiple alleles, have been found. These mutants have partly been isolated in the present mutation material and have partly been obtained from other collections of *Vicia faba* (see Table 4). With one exception, the flower colours of the different mutants in the *dp-a* locus are very similar to the colours found in the lines of the standard material. The colour of the mutants includes the standard, wings, and keel. The most frequent colour is dark brown – a colour, that is characteristic for 14 of the alleles. Six of the alleles confer a solid violet colour, whereas the others elicit intermediate colours between these two extremes. The wing spots are not apparent in the mutants of this locus. However, because of the genetic relationships, and since the spots on the stipules are visible, the wing spots are probably present, but they are masked by the other colours in the mutants. *Dp-a*^{29} is distinguished from the other mutants by its solid yellow flowers. *Dp-a*^{12} and *dp-a*^{13} were isolated in the same M_2 family, and the same was true for *dp-a*^{14} and *dp-a*^{15}. *Dp-a*^{20} and *dp-a*^{21} were isolated in the same inbred line. *Dp-31* was isolated in the M_3 generation in 1969 and has not been analysed yet. The greyish brown colour of the flower is also recovered in the calyx, in the pedicels, and in the terminal part of the stem – characteristics which distinguish this mutant from the others belonging to the *dp* group.

Table 5. Dominance relationships for flower-colour mutants belonging to locus *dp-a*

Gene symbol	F_2-segregation		$\chi^2_{(3:1)}$
	Normal	Mutant	
dp-a^1	70	24	0.0142; $0.95 > P > 0.90$
dp-a^2	114	45	0.716 ; $0.50 > P > 0.30$
dp-a^3	39	13	
dp-a^4	70	27	0.416 ; $0.70 > P > 0.50$
dp-a^5	53	16	0.121 ; $0.80 > P > 0.70$
dp-a^6	73	12	5.368 ; $0.025 > P > 0.01$
dp-a^7	139	31	4.149 ; $0.05 > P > 0.025$
dp-a^8	76	22	0.351 ; $0.70 > P > 0.50$
dp-a^9	71	18	1.083 ; $0.30 > P > 0.20$
dp-a^{10}	123	31	1.948 ; $0.20 > P > 0.10$
dp-a^{11}	43	13	0.095 ; $0.80 > P > 0.70$
dp-a^{12}	76	23	0.165 ; $0.70 > P > 0.50$
dp-a^{13}	13	33	0.286 ; $0.70 > P > 0.50$
dp-a^{14}	128	44	0.0310; $0.90 > P > 0.80$
dp-a^{15}	59	20	0.0042; $0.95 > P > 0.90$
dp-a^{16}	62	21	0.0040; $0.95 > P > 0.90$
dp-a^{17}	225	75	
dp-a^{18}	77	28	0.031 ; $0.80 > P > 0.70$
dp-a^{19}	31	7	0.877 ; $0.50 > P > 0.30$
dp-a^{20}	15	4	0.158 ; $0.70 > P > 0.50$
dp-a^{21}	38	3	6.837 ; $0.01 > P > 0.005$
dp-a^{22}	42	22	3.000 ; $0.10 > P > 0.05$
dp-a^{24}	15	8	0.989 ; $0.30 > P > 0.20$
Σ	1672	516	2.343 ; $0.20 > P > 0.10$

The data in Table 5 show that the alleles examined in the *dp-a* locus are recessive to normal

flower colour. The deviations from the 3 : 1 ratio for some of the alleles are due to a deficit of mutant types. Such a deficit has been observed earlier in many investigations of induced mutations, and have been discussed, e.g., by GOTTSCHALK (1961).

As mentioned earlier, the mutant character, or phenotype, was expressed in the F_1 progenies when two different allelic forms were crossed with each other. It may be expected that if the two parents in a cross have the same flower colour, the F_1 generation will display the parental phenotype for flower colour; and conversely, if the parents have different coloured flowers, the F_1 offspring will exhibit an intermediate flower colour. This was also the observation in most instances, but deviations from expected flower colour in some crosses – and especially the apparently irregular F_2 data – indicate a more complex inheritance of these colours. The alleles of this *dp-a* locus require a more exhaustive analysis, and at present the results that are available will be only summarized. For a brief orientation, the mutants are divided into dark brown, violet, and intermediate types. Within each of these groups, particularly in the last one, there is, however, a certain variation in the intensity of the flower colours among the F_1 progenies (see below) and particularly between plants in the various F_2 populations. Whether this variation is environmentally or genetically determined is presently unknown.

The dark brown to black colour, which is putatively imparted by melanin (ROWLANDS et al. 1962), seems to have a stronger penetrance than the other colours. In 51 dark brown × dark brown crosses, both the F_1 and F_2 generations mostly displayed dark brown corolla colour. In a few cases, the F_2 generation also segregated in types with a pink standard and dark brown wings. In five crosses, violet × violet, the F_1 offspring were violet, except in one instance, and then the standard was pink. The F_2 segregated in violet as well as in various intermediate types. When intermediate types were crossed, the same types were, as expected, obtained both in the F_1 and the F_2 generations. The composition of the colours in the different floral parts was different when the corolla colours of the parents were compare with the F_1 and F_2 progenies. Further, true violet types were also observed in the F_2 generation.

When dark brown types were crossed with violet ones, or reciprocally, the F_1 offspring had violet or pink standards and mostly dark brown wings, or in a few cases violet wings. The F_2 generation segregated in all possible types from dark brown to violet in apparently irregular frequencies. The same results were obtained after crossing dark brown types with intermediate types.

Violet flowers crossed with intermediate-coloured flowers gave rise to F_1 types with pink to violet standards and violet wings. The F_2 segregated in speciously irregular frequencies of all types; thus, in a few cases, even plants with dark brown flowers. The yellow phenotype has only been crossed with dark brown types. In this cross, the F_1 generation displayed dark brown flowers while the F_2 offspring included plants with both dark brown and yellow flowers.

In crosses between representatives of loci *dp-a* and *sp-a* or *sp-b*, the flowers of the F_1 progenies were normal. In the F_1, both crosses yielded a segregation close to a 9 : 3 : 4 ratio. In the F_2 of the cross *dp-a* × *sp-a*, 28 normal, 6 coloured, and 15 white-flowered plants were recovered ($\chi^2 = 1.732$; $0.50 > P > 0.30$). The F_2 of the cross *dp-a* × *sp-b* segregated with 60 normal, 13 coloured, and 25 white-flowered plants ($\chi^2 = 2.013$; $0.50 > P > 0.30$). These segregation ratios consequently indicate a complementary interaction of alleles from the *dp-a* locus and either the *sp-a* or the *sp-b* locus, where white can be regarded as epistatic to coloured, yielding a white, double-recessive offspring. The same ratio was obtained by BOND (pers. comm.) when *sp-a*^{1} was crossed with *dp-a*.

Only limited data are available for the mutant with yellow spots (*dp-b*^{1}). However, when crossed with the standard variety Primus, the F_1 generation was comprised of so-called normal offspring, whereas in the F_2 a segregation of 22 normal : 7 mutants was obtained, which indicated a monofactorial inheritance ($\chi^2 = 0.0115$; $0.95 > P > 0.90$). When *dp-b*^{1} was crossed with *dp-a*, the F_1 was normal and the F_2 segregated in 52 normal : 26 yellow-spotted : 27 coloured corollas. The yellow corolla colour of the above 26 plants was also expressed in a weaker intensity throughout the remaining floral parts. Thus, the phenotype was a light yellow flower with intense coloured yellow wing spots. The F_2 ratio coincided with a 9 : 3 : 4 Mendelian ratio in which the coloured

phenotype was probably epistatic to yellow wing spots. The same relationship was indicated when *dp-b*[1] was crossed with *sp-b*. The very limited data (9 normal : 4 white : 2 yellow-spotted flowers) indicate complementary gene interaction, where white is epistatic to yellow wing spots.

3. Types with modified seed-coat colour (*Sc, V, r, y*)

The typical colour of the seed-coat, or testa, in most cultivated varieties of *Vicia faba* in Europe is pale buff to buff. Wild forms of this species, for the most part, have black seeds, which – together with the fact that mutations from recessive to dominant are very uncommon (NILAN 1967) – indicates that the black seed colour most likely is the original one.

At Svalöf, chiefly four types with deviating seed-coat colour – green, violet, red, and black – are under study. At present, five types with green, seven with violet, and two with a red seed-coat have been assembled (Table 6). Of these, all but one have been obtained from different foreign collections. A violet type was isolated, however, in Svalöf in EI-treated material. Out of 43 types with a black seed-coat, 15 have been obtained from other collections (chiefly from Gatersleben), with most of them being wild forms native to the Balkans. Eighteen types have been isolated in X-irradiated material, eight in EI-treated, and two in MMS-treated material. It is extremely difficult to determine whether these 28 types and the previously mentioned violet type are true mutations or merely products of outcrossing in the generation before the respective treatments. Three of the isolations (*Sc-41*, *Sc-42*, and *Sc-43*) were made in the M_2 generation in 1968. At that time, the seeds that were used for treatments had been obtained from material that had been isolated in the greenhouse for several generations during which time no black-seeded types were observed. On the other hand, black-seeded types have been studied at Svalöf, and under such condition the chance of outcrossing can thus not be excluded. However, measures to preclude this contingency were taken, and these three mentioned types are regarded to be true mutations from the recessive to the dominant state.

Seed-coat colour is one of the best investigated characters in *Vicia faba* (KAZNOWSKI 1923; ERITH 1930; HEUSER 1931; SIRKS 1931; HUA 1944;

Table 6. Origin of types with changed seed-coat colour in *Vicia faba*

Gene symbol	Origin
Black-seeded	
Sc1	Isolated in X-irradiated material, 1956
Sc-2	Isolated in X-irradiated material, 1956
Sc-3	Isolated in EI-treated material, 0.125%, 1958
Sc-4-Sc-7	Isolated in X-irradiated material, 1956
Sc-8	Isolated in X-irradiated material, 4000 r, 1956
*Sc-9-Sc*13	Isolated in X-irradiated material, 1956
Sc-14	Isolated in X-irradiated material, 8000 r, 1956
Sc-15	Isolated in X-irradiated material, 4000 r, 1956
Sc-16	Isolated in EI-treated material, 0.05%, 1960
Sc-17	Isolated in X-irradiated material, 1957
Sc-18	Wild form from Italy, obtained from Gatersleben, 1961
Sc-19-Sc28	Wild forms from the Balkans, obtained from Gatersleben, 1961
Sc-29	Kiemes Schwarze Pferdebohne, obtained from Gatersleben, 1961
Sc-30	"Fioletowy", obtained from Warsaw, 1961
Sc-31	"Féverole de Florence", obtained from Rabat, 1962
Sc-32	Isolated in EI-treated material, 0.025%, 1959
Sc-33	Isolated in EI-treated material, 0.05%, 1959
Sc-34	Isolated in EI-treated material, 0.1%, 1959
Sc-35	Isolated in EI-treated material, 0.2%, 1959
Sc-36	Isolated in X-irradiated material, 6000 r, 1959
Sc-37	Isolated in EI-treated material, 0.0015%, 1959
Sc-38	Isolated in X-irradiated material, 8000 r, 1959
Sc-39	Isolated in EI-treated material, 0.025%, 1959
Sc-40	Obtained from Bot. Garden, Moscow, 1959
Sc-41	Isolated in X-irradiated material, 6000 r, 1967
Sc-42	Isolated in MMS-treated material, 0.02%, 1967
Sc-43	Isolated in MMS-treated material, 0.01%, 1967
Violet-seeded	
V-1	Isolated in Primus, 1959
V-2	Isolated in EI-treated material, 0.05%, 1960
V-3	Isolated in Primus, 1959
V-4	Obtained from Bot. Garden, Moscow, 1962
V-5	Obtained from Bot. Garden, Moscow, 1962
V-6	Obtained from ROWLANDS (Ad 99), 1964
V-7	Obtained from BRYSSINE, Rabat, 1964

(Table 6 continued)

Gene symbol	Origin
Green-seeded	
y-1	Wild form from China, obtained from Gatersleben, 1961
y-2	Obtained from ROWLANDS (Ch 193), 1964
y-3	From Japan through FAO, 1965
y-4	Obtained from BOND, Linie 349, 1968
y-5	Obtained from BOND, "Staygreen", 1968
Red-seeded	
r-1	Obtained from ROWLANDS (Ad 96), 1964
r-2	Obtained from PICARD (D 1434), 1967

and PICARD 1963). The data obtained in the different investigations are mostly uniform. Thus, the green and red colours were found to be recessive to buff, whereas black seed colour was dominant to buff. Examinations of two of the green-seeded lines (*y-1* and *y-2*) and 19 of the black-seeded lines (*Sc-1* – *Sc-15*, *Sc-17*, *Sc-18*, *Sc-28*, and *Sc-30*) in the present material gave the same relationships as described above. The two green-seeded lines segregated in 60 buff; 30 green ($\chi^2 = 3.333$; $0.10 > P > 0.05$), while in the black-seeded types, there was a deficit of plants with buff seeds within most of the lines. This deficit caused a significant deviation from the 3 : 1 ratio for the compiled data of the *Sc*-crosses (1148 black : 299 buff; $\chi^2 = 14.329$; $P < 0.001$).

A corresponding analysis of four types with violet seed-coats showed a violet F_1 and a F_2 generation that segregated in a 3 : 1 ratio (57 violet : 17 buff; $\chi^2 = 0.162$; $0.70 > P > 0.50$). Thus, violet was dominant over buff, which also was found by KAZNOWSKI (1923), ERITH (1930), HEUSER (1931), and PICARD (1963). In contrast to these data, SIRKS (1931) found buff colour to be dominant over violet in the line that he investigated.

The inheritance of two other seed colours has been investigated by PICARD (1963) and SIRKS (1931). PICARD found red to be recessive to buff. SIRKS analysed the seed-coat colour of plants with white flowers. As pointed out earlier, he found that the same factor, which he designated *O*, conferred the black spots on the wings and stipules, as well as the pigmentation of the corolla and seed-coat. The cinereous seed-coat colour in white-flowered plants was found to be recessive to the buff colour typical for most cultivated varieties including Primus. The *O*-factor also affects the violet, black, and probably even the red colours, since they can be expressed only when the *O*-factor is present in the dominant state. SIRKS denoted the genes for these above colours as cryptomeric in relation to the *O*-factor, which is comparable to hypostatic, or the *O*-factor is epistatic to the *S*, *V*, and *r* genes.

The allelic relationships within the different group of seed-coat colour types in the present material have not been investigated yet.

4. Types with altered pollen-grain shape (*po*)

A pollen grain in a typical plant is tricolpate, ellipsoidal, and about 1.7 times longer than broad (Fig. 2a). Three mutant types with modified pollen-grain shapes have been recovered. They have received the designations *po-1*, *po-2*, and *po-3*. *Po-1* was induced by X-irradiation, 4000 r in 1960; *po-2* by neutrons, 140 rad in 1962; and *po-3* was induced with MMS, 0.010 per cent in 1968. *Po-1* is characterized by having mostly triangular-shaped pollen when observed in slide preparations (see Fig. 2b). The pollen of *po-1* is about the same length as the pollen in the standard type, but are broader at the base of the "triangle".

Po-2 has pollen grains that appear to be spherical in shape as seen in the slide preparations and as compared with *po-1*; but the pollen of *po-2* is clearly distinguishable from the pollen of the standard material, which is the typical form for the species (see Fig. 2c). Moreover, the pollen of *po-2* is not so long nor as broad as the typical, or "standard", pollen, as characterized by the pollen of Primus. Finally, *po-3* is almost intermediate in appearance between *po-1* and *po-2*, but somewhat more like *po-2*. The size of the pollen is about the same as the pollen in Primus.

Pollen viability, based on pollen fertility, is somewhat reduced in these mutants. From an average of measurements from 3 different years, *po-1* exhibited a pollen fertility of 73 per cent, whereas *po-2* had 81 per cent fertility and *po-3*, 70 per cent pollen fertility. *Po-2* and *po-3* were measured in 2 consecutive years. The pollen fertility of Primus was not determined in the present investigation, but was found to be 97 per cent previously.

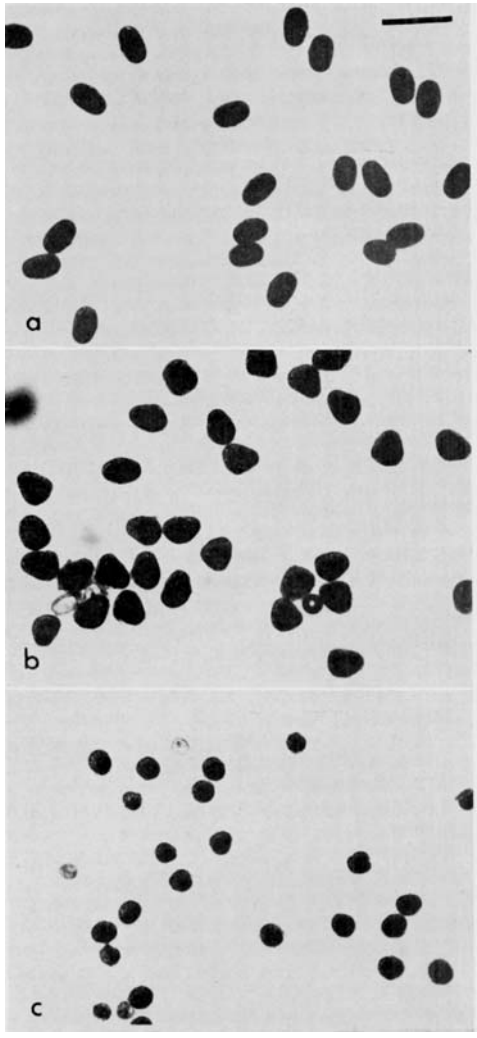


Fig. 2. Two mutants of *Vicia faba* with altered pollen-grain shape; a: the parental variety; b: the mutant *po-1*, pollen grains characterized by their triangular shape; c: spherical-shaped pollen grains of the mutant *po-2*.

Po-1 has been tested for dominance relationships and was found to be recessive to the standard material. In the cross *po-2*×normal (Primus), the F_1 was normal phenotypically, but in the F_2 generation a segregation of 115 normal phenotypes: 28 mutants was recovered ($\chi^2 = 2.240$; $0.20 > P > 0.10$). The fact that the F_1, as well as the 115 normal plants in the F_2 generation did not segregate with regard to the shape of the pollen grains indicates that this character is determined by the genotype of the mother plant. A parallel case in *Lathyrus odoratus* was described by BATESON (1913) who found that long pollen shape was dominant to round pollen shape.

5. Types with a terminal inflorescence (*ti*)

A standard plant of *Vicia faba* is characterized by indeterminate growth, which normally ends when a maximum number of pods have been produced and, of course, when the plant has reached a certain age. Two mutants have been isolated on different occasions, but both exhibit the same deviation from the standard material. One of the mutants was induced in 1962 (neutrons, 35 rad) and the other in 1968 (MMS, 0.015 %). The genes are designated *ti-1* and *ti-2*.

The mutant types are characterized by a terminal inflorescence. The development is normal until the plants reach the third or fourth inflorescence stage. At this time, the vegetative growth is terminated and the stem is transformed into a peduncle subtending an inflorescence (see Fig. 3). Rather soon, a few lateral shoots develop from the lower leaf axils, and at the same time generally one to a few lateral shoots arise from the base of the stem. Both types of lateral shoots exhibit the same abnormal development as the primary stem. As can be seen in Fig. 3, the peduncle of the inflorescence that terminates the stem is thicker and larger than normal, i.e., it is intermediate between the normal stem and peduncle. The apical flower in the uppermost inflorescence often is similar to the cauliflower-head type observed in unifoliate mutants and described in detail by SCHEIBE and GOTTSCHALK (1956). As expected, this mutant type is shorter than the standard type, which is illustrated in Fig. 3b.

The dominance relationships were tested for one of the mutants (*ti-1*). The character displayed monofactorial inheritance in which the F_1 was normal and the F_2 segregated in 90 normal phenotypes: 23 mutants ($\chi^2 = 1.301$; $0.30 > P > 0.20$).

At the time when the mutant was isolated, it was considered as an interesting curiosity. During recent years, it has, however, attracted a great deal of interest among plant breeders and others engaged in crop production of *Vicia faba*. This interest is based on the good possibilities of



Fig. 3. A mutant of *Vicia faba* with a terminal inflorescence; a: at the flowering stage; b: at the seed maturation stage. At the left, the parental variety; at the right, the mutant.

eliminating some of the disadvantageous features in the standard cultivated varieties grown to-day by using this mutant type in the breeding program. As is seen in Fig. 3b, the pods in a normal plant are distributed along the stem in the lower half of the plant, with the lowest pods rather close to the soil surface. The distance from the base of the plant to the first pod, or more correctly, the node number of the first inflorescence, is correlated with earliness. Since the lateral shoots, as well as the main shoot, in the mutant with a terminal inflorescence reach about the same height level, the pods likewise will be correspondingly at about one level. This is a positive attribute when combines are used for threshing.

The upper half of the plant, above the level of the pods, as in the standard material, naturally, has a delaying effect on the ripening, or maturing, process of the pods. This is, indeed, a great disadvantage owing to the long vegetative period that is characteristic of *Vicia faba*, and is particularly relevant in years when the critical months – in Sweden, August, September, and even up to November – have high precipitation. By transferring the mutant character (*ti*) to highly bred cultivated varieties, it would be possible to elim-

Table 7. Number of inflorescences on plants with a terminal inflorescence before and after the transfer of the *ti*-gene to new genetic environments

Stage	Number of inflorescences					
	3	4	5	6	x	n
Before crossing	18	30			3.6	48
After 3 generations of outcrossing	22	48	28	2	4.1	100

inate the upper part of the plant and as a result achieve a more uniform and earlier ripening.

At the present time, however, there are certain reservations as regards the usefulness of this character. First, it is not known yet if the removal of the upper vegetative portion of the plant will have a negative effect on the total photosynthetic production, and thus on seed weight. Secondly, it would, of course, be desirable if the growth of the plant was terminated at the level where the last pod normally is formed in a standard plant. Presently, the growth terminates, as mentioned earlier, at the level of the third or fourth inflorescence, whereas pods are normally produced up to about the seventh to tenth inflorescence. It is known, however, that the character expression of definite genes can be changed by changing the genetic environment (LOESCH 1964; GAUL et al. 1968). It would therefore, perhaps, be possible to alter the point of growth termination by transferring this mutant gene to new genetic environments. Thus, the *ti-1* gene was crossed to variety Primus by mixing seeds of the mutant and Primus. This mixture was sown outdoors and harvested for three consecutive generations. Measurements were made of the number of inflorescences before crossing and after the third generation (Table 7). It was then found that the changed expression was in the desired direction. The average number of inflorescences was increased from 3.6 to 4.1, and about one-third of the plants in the third generation had 5 or 6 inflorescences – a number which was not found in the original mutant plants. It may be possible to obtain a still more pronounced change in the expression of the *ti* gene by crossing it to cultivated varieties less closely related to Primus. Such crosses have been made.

Although the growth of the mutant plant terminates at the fourth inflorescence and as a consequence produces a reduced number of pods on the main stem, this is compensated for, to a certain degree, by the pod production of the lateral, or axillary, shoots. The mutant will be tested in yield trials for the first time in 1971.

6. Early mutants (*ea*)

The first early flowering and early ripening mutant that was found in the present mutation material (*ea-1*) was isolated in the M_3 generation of an X-irradiated family (7000 r) in 1961. Since then, 14 additional mutants of the same type have been isolated in material treated with neutrons, gamma-rays, EMS, MMS, and EMS+Cu^{++}. Only *ea-1* has been investigated further. This mutant is included in the breeding material of *Vicia faba* at Svalöf and has been registered as Sv 0420.

The morphological examination of Sv 0420 constitutes part of an investigation of the differences in optimum conditions for growing early types versus late types. It will be published later in full detail. The main part of the data presented here was collected by O. Ojomo, Nigeria.

Sv 0420 is characterized by its reduced stature and more slender growth as compared with Primus. The slender growth habit of Sv 0420 has not been confirmed statistically. However, this character has been analysed for Primus and another early variety, Pirhonen. Significant differences were found in leaf area and fresh and dry weights. Pirhonen, an old Finnish variety, is earlier than Sv 0420 and has a more slender growth habit, which implies that if a corresponding comparison had been made between Primus and Sv 0420, the differences would probably have been less accentuated.

Plant height was measured in both Primus and Sv 0420 (Table 8). The mutant was significantly shorter than Primus. The short stature seems to be due to a reduction in the number of nodes, a characteristic that was also significantly lower than in Primus (Table 8). The internode lengths were not measured. The data in Table 8 for this character were obtained by dividing the number of nodes by the plant height. The values obtained indicate that the difference in plant height between Primus and Sv 0420 is mainly due to a reduced number of nodes in the mutant.

In *Vicia faba*, earliness can be measured in

Table 8. Measurements of some morphological characters in Primus and Sv 0420

Character	Primus	Sv 0420	t-value
Node of first inflorescence	8.45 ± 0.10	6.32 ± 0.11	13.51***
Plant height (cm)	132.53 ± 1.07	110.67 ± 1.71	13.10***
Node number	27.48 ± 0.36	23.35 ± 0.33	12.52***
Internode length	4.8	4.7	

three different ways: (1) by the node at which the first inflorescence appears; (2) the number of days from sowing to the opening of the first flower; and (3) the number of days from sowing to pod ripening. Of these three characters, the last-mentioned one is influenced most by environmental conditions and therefore is less reliable. On the other hand, this is the most important character from the practical point of view. The node at which the first inflorescence appears is found to be closely correlated with earliness (e.g., GOTTSCHALK 1960; BOND and FYFE 1962) and is commonly used as a selection index in plant breeding. The first inflorescence in Sv 0420 appears at a significantly lower node as compared with Primus (Table 8).

Comparative trials carried out at Svalöf and branch experiment stations in southern and central Sweden (see Table 9) show a difference of 4 to 5 days in earliness between Primus and Sv 0420, that is, measured as number of days from sowing to the opening of the first flower and also from sowing to ripening. The difference in earliness is somewhat less than what was observed under greenhouse conditions and in small field plots during previous years. In these investigations, Sv 0420 was found to be 6 to 7 days earlier than Primus. However, the data presented in Table 8 are based on trials from the years 1968, 1969, and 1970; and these years were unfavourable for the early type. In 1968 and 1969, the weather conditions were very favourable during the period from anthesis to pod maturity. Hence, late varieties ripened more rapidly than usual, and the differences in earliness were offset. The past year (1970) was characterized by high precipitation, which resulted in a prolonged vegetative period, a circumstance that should have given a maximum manifestation of differences in earliness. But this manifestation was lost because of a few days of subzero temperatures between the ripening of Sv 0420 and Primus. This caused an artificial ripening of the pods in Primus, resulting in an earliness difference that was less than expected.

The usual seed rate for *Vicia faba* varieties that are cultivated in Sweden is around 200 kg per hectare. Under these conditions, the yield of Sv 0420 is about 10 per cent lower than the yield of of Primus (Table 9). Considering its slender growth, Sv 0420 can, however, be regarded as a specific type that may need very different growing conditions to produce an optimum yield. Thus, for instance, owing to its growth habit, a field stand of Sv 0420 will be less dense than one of Primus if the same seed rate is used. A certain increase in the number of plants per hectare will be obtained in Sv 0420 as a result of its lower thousand-kernel weight, but it is obvious that this is not enough to give a yield comparable to Pri-

Table 9. The effect of different seed rates on yield in Primus and Sv 0420

Variety and seed rate		Number of trials	Yield		Thousand-kernel weight g	Days from sowing to:		Plant-height cm
			Kg/ha	rel.		flowering	ripening	
Primus	200 kg/ha	9	3525	*100*	384	56	137	109
	250 kg/ha	9	3759	107	391	56	136	107
Sv 0420	200 kg/ha	9	3158	90	346	52	132	91
	250 kg/ha	9	3558	101	350	51	132	89

Table 10. Data from comparative trials with early ripening populations

Variety	Number of trials	Seed yield		Thousand-kernel weight g	Days from sowing to:		Protein content		
		Kg/ha	rel.		flowering	ripening	%	Kg/ha	rel.
Primus	3	5634	*100*	428	56	139	31.05	1749	100
Sv 0420	3	5014	89	373	52	138	29.76	1719	90
Sv 0680 (*Sc-8* × *V-3*)	2	4867	87	350	48	126	31.77	1546	89
Sv 0700 (*Sc-7* × *Sc-12*)	3	4645	82	307	50	127	33.63	1562	89
Sv 0720 (*Sc-9* × *Sc-12*)	3	3846	68	279	49	127	35.74	1375	79
Sv 0740 (*Sc-12* × *Sc-8*)	3	4522	80	269	50	123	32.96	1490	85
Sv 0760 (*Fr-38* × *Sc-7*)	2	4264	90	312	53	127	31.74	1353	85

mus. Thus, it seems to be necessary to increase the seed rate of Sv 0420 if an optimum yield is to be obtained; and it applies in particular if a yield at the same level as for Primus is to be reached. This has been tested in the trials described in Table 9 by sowing both cultivars at different seed rates. From this table, it is seen that when Sv 0420 is sown at 250 kg per ha., it reaches at least the same yield as is obtained when Primus is sown at 200 kg per ha.

In Table 9 it is also evident that the yield of Primus increased when the seed rate was increased from 200 kg to 250 kg per ha. The results seem to signify, then, that the generally accepted seed rate of 200 kg per ha. is not the optimum one. Furthermore, the optimum seed rate for Sv 0420 must be still higher when considering the yield gain in Primus at a sowing rate of 250 kg per ha. and the differences in the growth habit between it and Primus.

Primus ripens for the most part too late north of the three southernmost provinces of Sweden – Skåne, Blekinge, and Halland. On the other hand, in the three southern provinces, the yield is satisfactory, and thus economically profitable. To extend the limit of cultivation to central Sweden, it is necessary to introduce an earlier variety. The important comparison in this connection is therefore between Primus, 200 kg per ha., and Sv 0420, 250 kg per ha. (Table 9). Under the assumption that the present cultivation of Primus is acceptable, the yield figures in Table 9 indicate that if the growing conditions are adapted to the early type, then it is also possible to cultivate *Vicia faba* in central Sweden with insured economic return.

The data obtained from very preliminary studies of the inheritance of the *ea-1* type are contradictory and unreliable because the earliness of the F_1 generation cultivated under greenhouse conditions was estimated on the node number of the first inflorescence, whereas the earliness of the F_1 generation grown under field conditions was estimated when the plants were mature. In addition, the number of individuals were limited. The F_2 exhibited the first inflorescence at the fourth, fifth, and sixth nodes, which would indicate an incomplete dominance for earliness, since *ea-1* that year formed its first inflorescence at the fourth or fifth node. The F_2 generation segregated in 29 normal maturating plants : 10 early maturating plants, thus indicating that *ea-1* is recessive to the normal type, or here, Primus. No additional comments are justified now. The investigation will be repeated in greater scale under similar conditions.

When testing the allelic relationships between the types with deviating seed-coat colour, it was observed that some of the F_2 combinations matured very early. The parents (*Sc-7*, *Sc-8*, *Sc-9*, *Sc-12*, and one other, *Fr-38*) ripened 6 to 7 days earlier than the standard variety under greenhouse conditions. The early maturing F_2 recombinant types were selected and multiplied. They have now been tested in two or three years of comparative trials at Svalöf. The results from these trials are tabulated in Table 10. From the table, it is apparent that the populations are 3 to 8 days earlier at the time of anthesis. This difference is accentuated at the end of the vegetative period, where the early populations – in spite of that these trials were performed during the 3 unfavourable years discussed above – were 12 to 16 days earlier in maturing than Primus.

As is also apparent from Table 10, the yields

were lower for these mutant populations when compared with Primus. The seed rate was, however, 200 kg per ha. With regard to what has been discussed earlier about optimum conditions, together with the fact that these mutant populations, except one, yield about the same as Sv 0420 (cf. Table 10), it is probable that the mutant populations will yield as much as Primus if the optimum growing conditions can be provided.

7. Population with high protein content

Among the F_2 populations from crosses between seed-coat colour types described previously and presented in Table 10, one population displayed a higher protein content when compared with other populations. This population, derived from the cross *Sc-9* × *Sc-12*, is included in the breeding material at Svalöf and is registered as Sv 0720. As an average of three years of trials, Sv 0720 had a protein content of 4.69 per cent units higher than Primus and 2.11 per cent units higher than the best of the other populations. The increase over Primus is equal to 15.1 per cent (Table 10).

Comparable values for the parents, *Sc-9* and *Sc-12*, are not available because they have only been grown under greenhouse conditions. The average values of protein content, based on 4 to 6 generations under these conditions, were 34.3 per cent for *Sc-9* and 32.7 per cent for *Sc-12*, and 31.7 per cent for Primus, which was also grown under greenhouse conditions. Thus, *Sc-9* displays an extreme protein content, but the seed samples from each generation have been small and the data are therefore somewhat unreliable.

Sc-9 and *Sc-12* were isolated in X-irradiated material as types with altered seed-coat colour. They were also earlier than the parental variety and had a protein content that was higher. When the mutants were combined, the F_2 and later generations exhibited – besides the seed-coat character, which has been shown to be caused by major genes – a more extreme earliness and an increase in protein content as compared with the parental mutants. It is impossible to ascertain if the irradiation has evoked mutations in genes with pleiotropic effects or if mutations have been simultaneously induced in several genes affecting the two characters. It is, however, more probable that the second possiblility is germane with regard to indications that earliness is due to major gene action, whereas protein content is polygenically inherited. Since *Vicia faba* is a cross-fertilizer, it cannot be ruled out that the two mutants are extreme types within the latitude of variation of protein content in the parental variety, and which by chance happened to be isolated.

Individual plants from Sv 0720 were tested in co-operation with MUNCK and LÖFQVIST for protein content in 1969. In 277 plants that were tested, the variation was between 26 and 36 per cent. The 10 per cent breadth of variation was of the same magnitude of that found in two other populations (Pirhonen and Sv 0400). Pirhonen varied between 23 and 33 per cent and Sv 0400 between 23 and 34 per cent. In all three populations, plus selection was carried out. All plants with more than 33 per cent protein were selected in Sv 0720, whereas in the other two populations all plants with more than 30 per cent protein were selected. In addition, a minus selection was made in Sv 0720 in which all plants with less than 29 per cent protein content were selected. One hundred plants from each of the selections were retested in 1970, and the compiled data for the two selections in Sv 0720 are projected in Fig. 4. The variation was now between 28.5 and 39.3 per cent for the plus selection and between 21.7 and 35.5 per cent for the minus selection. The average values for the two selections were 33.7 ± 0.20 and 31.1 ± 0.23 per cent, respectively. The differences between the two average values are significant at the 0.1 per cent level. The variation for the plus selections in the other two populations were 27 to 35 per cent for Sv 0400 and 26 to 38 per cent for Pirhonen, and the average values were 31.0 and 31.5 per cent, respectively. Thus, the two average values were not higher than the average values for the minus selection in Sv 0720.

The data described here show that Sv 0720 has a higher protein content than the other two populations and that selection in Sv 0720 obviously has outstanding possibilities to yield a positive result because the prompt response of selection indicates high heritability for protein content.

At present, it is not known whether the quality of the protein also is altered in Sv 0720. The economic value of Sv 0720 would, of course, increase if it showed a positive change especially in the amount of methionine, the primary limiting amino acid in *Vicia faba*, as well as in most Leguminosae. Since Sv 0720, and the other two populations as well, had a lower thousand-kernel weight than, e.g., Primus, the correlation between

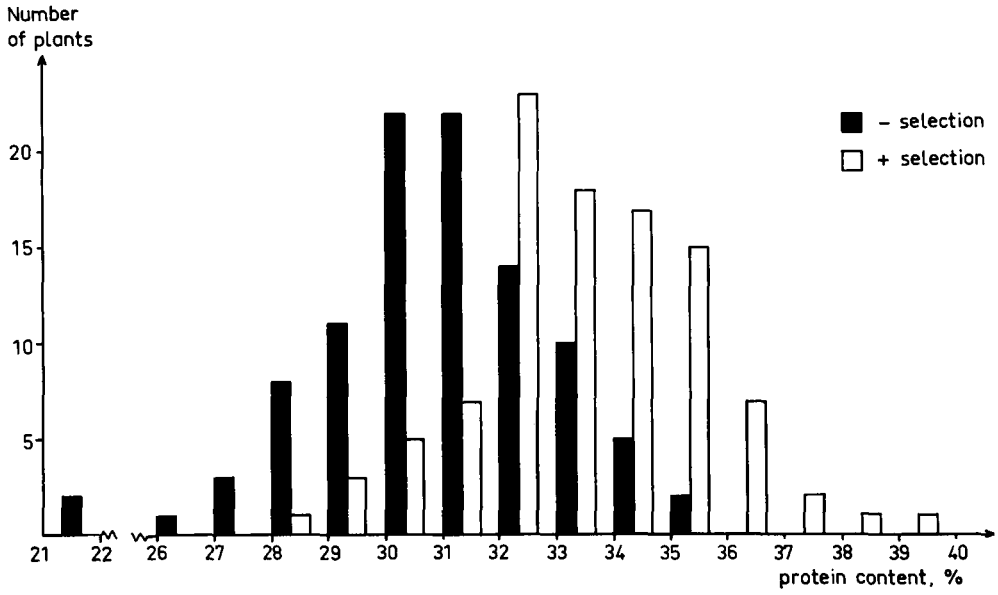


Fig. 4. Distribution of plants with different protein content. The plants are derived from one year's plus and minus selections in Sv 0720, a population with high protein content.

this character and the protein content was checked to determine if the high protein content was due to a reduced thousand-kernel weight; but there was no correlation. On the contrary, there was an indication of a positive correlation between thousand-kernel weight and protein content. The correlation coefficients were +0.37 for Sv 0720 and +0.24 and +0.22 for the other two populations.

8. Other mutation types

For the sake of completeness and as a source of information of other available material, the remaining mutation types that have been isolated in the present material will be briefly described, but only briefly since the investigation of these types is presently very limited or they have been discussed elsewhere (SJÖDIN 1962).

A. *Red-stemmed seedling* (Rs)

This character is most easily observed in the seedling stage. The stem is then distinctly red-pigmented – a pigmentation that is purportedly due to the presence of antocyanins (ROWLANDS et al. 1962). The strength of the colour regresses during the development of the plant. At the flowering stage, the colour is just perceptible, with the pigmented region now limited to the uppermost 5–10 cm of the plant. The character is dominant to the normal green stem colour. The F_1 generation was red-stemmed and the F_2 segregated in 80 red-stemmed seedlings: 23 normal seedlings ($\chi^2 = 0.392$: $0.50 > P > 0.30$).

B. *Dwarfs* (dw)

One dwarf type was obtained from BOND in 1964. At the stage of pod maturity, the height of the dwarf mutant is about 1.5 dm. The character is recessive to normal plant height and segregated in 32 normal plants: 3 dwarfs, thus exhibiting a considerable deficit of the mutant type ($\chi^2 = 5.037$: $0.025 > P > 0.01$). Four dwarf types have been isolated in the present material. They are extreme in their expression, and are at the flowering stage about 5–10 cm tall. The flowers do not develop and consequently no seeds are formed. The mutant thus had to be maintained in the heterozygous state. It has not yet been determined whether the dwarf character is due to a

reduced number of nodes or a reduction of internode length or both. From observations, it seems, however, probable, that a reduction of internode length is the main factor responsible for these dwarf types.

C. *Late ripening type* (Lf)

This type was isolated in X-irradiated material (10,000 r). It flowers and ripens about 15 days later than the standard material. The character is dominant over normal flowering time. The F_1 was late flowering and the F_2 generation segregated in 43 late flowering plants: 15 normal flowering plants ($\chi^2 = 0.023$; $0.90 > P > 0.80$).

D. *Hooked seed* (ho)

This mutant has a bulge on both cotyledons just above the embryo, which confers a characteristic hook to the seed. This oddity was isolated in EI-treated material (0.015 %) and displayed monofactorial inheritance. The F_1 generation was comprised of phenotypes like the normal parent, but the F_2 segregated in 13 normal phenotypes : 5 mutant plants ($\chi^2 = 0.0741$; $0.80 > P > 0.70$).

E. *Small hilum* (Hi-1 and Hi-2)

The hilum of a normal seed of the standard cultivated variety Primus is black, about 4–5 mm long and about 1 mm broad. Two types with a small hilum have been isolated (*Hi-1* and *Hi-2*). These two mutant types are phenotypically similar, with the hilum about 3 mm long and 0.50–0.75 mm broad. The F_1 generations from the crosses *Hi-1*× Primus and *Hi-2*× Primus displayed a small hilum, indicating that the small hilum allele is dominant to the normal allele. The F_2 generation has not been analysed yet for the second cross, whereas *Hi-1*× Primus segregated in 16 small hilum : 4 normal hilum plants ($\chi^2 = 0.267$; $0.70 > P > 0.50$).

F. *Spherical hilum* (Hi-3)

The shape of the hilum in this type is nearly spherical, but somewhat longer than broad. The F_1 of the cross *Hi-3*× Primus showed the spherical type, indicating that this character is dominant over normal shape.

G. *Long hilum* (Hi-4)

The hilum is about 6 mm long and 1 mm broad. The F_1 generation of the cross *Hi-4*× normal was comprised of plants with long hilum seed, which indicates that even this character is dominant over normal hilum shape. The F_2 generations have not been analysed for both *Hi-3* and *Hi-4*.

H. *Colourless hilum* (n)

Colourless hilum is a common deviation from the normally black pigmented hilum. The white, or non-pigmented, hilum is found in most populations of *Vicia faba*. It is generally recognized that hilum colour exhibits monofactorial inheritance with black dominant over colourless (ERITH 1930; SIRKS 1931; PICARD 1953; FYFE and BAILEY 1951; HOLDEN and BOND 1960). The type with colourless hilum that was isolated in the present material also shows the same inheritance. The F_1 from the cross *n*× normal (Primus) had a black hilum and the F_2 offspring segregated in 37 black hilum plants: 12 colourless hilum plants ($\chi^2 = 0.0068$; $0.98 > P > 0.95$).

I. *Chlorophyll mutations*

As in most material, chlorophyll mutations appear in different frequencies after all types of mutagenic treatments. In the present material, this frequency is about 2 to 3 per cent after treatments with different ionizing radiations and between 2 to 14 per cent after treatments with different chemical mutagens (SJÖDIN et al. 1968). It ought to be mentioned that these frequencies were obtained under field conditions. As described earlier, (SJÖDIN 1962), *viridis* is the most common type with *xantha* the next most common mutant type, whereas *albina* is a very rare mutant in *Vicia faba* as in most Leguminosae. A few *chlorotica variomaculata* types have also been observed. Only types that survive in the homozygous condition, have been isolated, and among them some have been tested for dominance relationships. As expected, they displayed monofactorial inheritance, with the normal type dominant to the chlorophyll mutants. In all crosses, there were deficits of the chlorophyll mutants in the F_2 generation.

J. *Asynaptic mutants* (as)

Eleven asynaptic mutants have been isolated. The mode of origin and the behaviour of these

mutants have been described in detail earlier (SJÖDIN 1970). The only thing that will be mentioned here is the designation of these mutants. According to the nomenclature used in this paper, the mutants will be discussed in the future employing the following symbols: *as-1* (A2), *as-2* (A8), *as-3* (A10), *as-4* (A11), *as-5* (A12), *as-6* (A13), *as-7* (A16), *as-8* (A20), *as-9* (A21), *as-10* (A22), and *as-11* (A35). The symbols used in the previous paper are indicated in parentheses.

Discussion of genetic data

Most information available about inheritance of different characters in *Vicia faba* derives its origin from a single paper by SIRKS (1931). He investigated 26 factors and in addition calculated the recombination frequencies between most of these genetic factors (genes). When grouping the values obtained based on the degrees of linkage, SIRKS found that 19 out of the 26 factors formed four linkage groups. It should be pointed out in this respect that the basic chromosome number of *Vicia faba* is 6.

It would, of course, have been very interesting to have had the opportunity to use SIRKS' extensive material in order to combine the four genetic linkage groups with the chromosomes cytologically. This is unfortunately impossible since all of SIRKS' material was lost during the Second World War. From the description of the characters in SIRKS' paper, it may, however, be assumed that some of his characters are identical with the gene markers available in the present collection and in other existing collections of *Vicia faba* types.

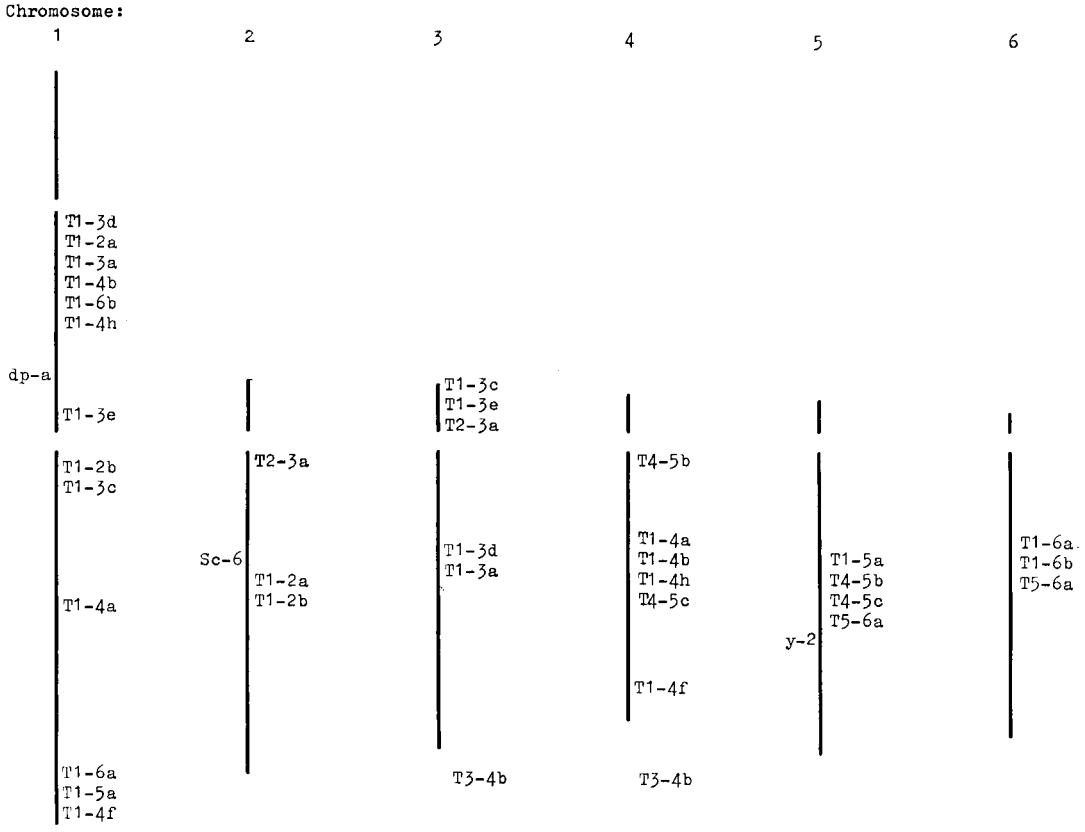


Fig. 5. Idiogram of *Vicia faba* showing the translocation break points used in the linkage studies of the genes *dp-a*, *Sc-6*, and *y-2*.

The basic material for locating the marker genes in their respective chromosomes is now available. About 100 translocation lines have been identified with regard to the chromosome involved and the approximate position of the break points (SJÖDIN 1971 b). At present, however, the linkage studies are restricted to three characters that are identified to chromosomes. The translocations used and the putative location of the genes are illustrated in Fig. 5 and the obtained recombination frequencies are presented in Tables 11, 12, and 13. From these data, it seems probable that the *dp-a* locus (the flower-colour locus) is situated in the short arm of chromosome 1, even if the data, to some extent, are contradictory (Table 11). Thus, one translocation in the long arm also manifests strong linkage with the *dp-a* locus (T1-2b).

One of the genes for green seed-coat colour (*y-2*) exhibited linkage with the T4-5c and T5-6a translocations (Table 12), indicating that this gene is located in the long arm of chromosome 5. The *Sc-6* gene (black seed-coat colour) displayed linkage with the T1-2b translocation, but showed free recombination with the other translocations presented in Table 13. It seems therefore probable that this gene is located in the long arm of chromosome 2.

BOND (pers. comm.) has found certain indications that *sp-a*^{1} (white flowers) is located either in chromosome 1 or in chromosome 3. However, when *sp* and *dp* are crossed, these genes segregate independently of each other, so it is less probable that *sp* is situated in the short arm of chromosome 1. BOND also found indications that the fertility restorer gene (*Df*) is situated in chromosome 2. It may be mentioned, too, that the *un-a* locus (unifoliate leaf) showed free recombination when crossed with T1-2a, T1-3a, T1-3c, T1-3e, T1-6a and T4-5b.

The most interesting of these preliminary linkage data is that the *y-2* gene seems to be situated in chromosome 5, since one of the genes in SIRKS' first linkage group is his *y*-gene. If this

Table 11. Data from F_1 of crosses (*dp-a*^{3} × translocation) × *dp-a*^{3}

Combination	χ^2_L	Recombination frequency	*n*
dp-a^3 × T4-5b	3.27	free	135
dp-a^3 × T1-6a	0.0123	free	81
dp-a^3 × T1-3a	20.78	26 ± 4.7	89
dp-a^3 × T1-3c	76.97	19 ± 2.8	203
dp-a^3 × T1-2a	24.47	31 ± 3.5	178
dp-a^3 × T1-3e	25.70	27 ± 4.0	122
dp-a^3 × T1-4h	73.92	10 ± 2.8	117
dp-a^3 × T1-5a	0.754	free	65
dp-a^3 × T1-4a	36.94	25 ± 5.4	65
dp-a^3 × T1-4f	0.871	free	93
dp-a^3 × T1-3d	4.83	41 ± 4.0	151
dp-a^3 × T1-2b	67.00	0	67
dp-a^3 × T1-4b	47.43	19 ± 3.5	125
dp-a^3 × T1-6b	42.38	12 ± 3.7	74

Table 12. Data from F_1 of crosses (*y-2* × translocation) × *y-2*

Combination	χ^2_L	Recombination frequency	*n*
y-2 × T1-6a	0.0720	free	125
y-2 × T3-4b	0.0345	free	116
y-2 × T1-2a	1.07	free	60
y-2 × T2-3a	1.14	free	43
y-2 × T4-5c	3.86	29 ± 9.9	21
y-2 × T5-6a	64.0	14 ± 3.2	123

Table 13. Data from F_1 of crosses (*Sc-6* × translocation) × translocation, as well as from F_2 of *Sc-6* × translocation

Combination	Backcrosses			F_2		
	χ^2_L	Recombination frequency	n	χ^2_L	Recombination frequency	n
Sc-6 × T1-6a	2.35	free	170	0.88	free	218
Sc-6 × T1-3a	0.0690	free	58	—	—	—
Sc-6 × T1-3c	0.532	free	188	—	—	—
Sc-6 × T1-2a	37.2	8 ± 3.7	52	64.6	7 ± 2.0	341
Sc-6 × T1-3e	0	free	220	2.25	free	314

gene for green seed-coat colour is similar to the *y-2* gene in the Svalöf collection, it would indicate that this genetic linkage group is correlated to chromosome 5.

With regard to the other three linkage groups of SIRKS, there are some characters that are similar to the ones available in the Svalöf collection. Thus, the *p*-gene (violet seed-coat colour) is situated in the second linkage group. However as described earlier, SIRKS found the violet colour to be recessive to buff, which is contrary to the data from the present investigation and also to that from other investigations (KAZNOWSKI 1923; ERITH 1930; HEUSER 1931; PICARD 1963). It is thus probable that SIRKS' p-gene is different from the ones in the Svalöf collection. The second linkage group also includes a gene for leather-like pods, and one for strong anthocyanin lines in the standard. Both characters have been observed in Primus, but have not been isolated as yet. The same is true for a gene *D* in the fourth linkage group, which produces a strongly developed felt indumentum inside the pod. These characters could probably be isolated, hence providing an opportunity to locate even these linkage groups with respect to the chromosomes cytologically. The third linkage group includes the gene for hilum colour (*N*), a character available in the present collection.

The unifoliate character has been observed in many species of the Leguminosae, which have been recently treated by GOTTSCHALK (1969). A few other cases are also known (BRYSSINE 1962; JESWANI et al. 1962; SANTOS 1969). A comparison of these mutant divulges several interesting similarities and dissimilarities. The venation of the single leaf clearly indicates that the unifoliate character in *Vicia faba* is a result of a fusion of the normal leaflets into a single lamina. The unifoliate character in some of the other mutants, e.g., in *Medicago sativa* (BAYLY and CRAIG 1962) and *Melilotus alba* (GOPLEN 1967), is the result of a suppression of the growth and development of the lateral leaflet primordia. This is also indicated through a comparison of venation between the single leaf and a standard leaflet. In those cases where the character has originated through a connation of leaflets, the unifoliate leaf usually has a larger leaf area as compared with any one of the leaflets in the typical leaf.

The unifoliate character behaves differently in different mutants with regard to the strength of expression. In some cases the character is expressed throughout ontogeny (obligate type), whereas in others there is a transition from unifoliate to the standard, or typical, type (transnormal). Spontaneous mutations seems to be of the transnormal type more often than induced mutations are. Out of 16 spontaneous mutations, 11 were of the transnormal type, whereas out of nine induced mutations, eight were of the obligate type. This type of mutation thus seems to be more pronounced in the induced mutations as compared with spontaneous ones.

Another interesting feature of these mutants is that besides the single leaf most of them have a modified floral structure, which leads to various degrees of sterility. Out of 25 known cases of unifoliate mutants, 21 also exhibited aberrant floral organs. One unifoliate mutant of *Trifolium repens* (ATWOOD 1938) did not form flowers. The most frequent floral modification in the unifoliate mutant is the so-called cauliflower-head inflorescence, in which the main difference is a compound branched inflorescence with rudimentary flowers as compared with the typical inflorescence (e.g., LAMPRECHT 1933, 1935; BAYLY and CRAIG 1962; GOPLEN 1967; SANTOS 1969). In the partially sterile unifoliate mutants of *Vicia faba*, the reduced fertility was mainly caused by a reduction of ovules and that sometimes the carpel was open, which permitted the ovules to dry out.

In sum, there are different causes for the expression of the unifoliate character, which, too, can be expressed in different strengths (obligate and transnormal types). Furthermore, during the formation of the flower, abnormalities of various kinds and degrees occur in one or more of the floral organs. These multiple effects involve different systems in the plant organization. According to GOTTSCHALK (1968), three interpretations explaining these multiple effects are possible: (1) A single mutant gene is responsible for the entire complex (true pleiotropic gene action). (2) A small portion of a chromosome has been lost containing several genes. (3) Several closely linked or neighbour-genes have mutated. As GOTTSCHALK (l.c.) claims, it is practically impossible in most cases to state which of these possibilities is realized. However, the two last interpretations would imply that for the known cases of unifoliate mutants in the Leguminosae (12 species in six genera) a gene affecting the

leaf-shape character should be close to one affecting floral organization in all the species that were investigated. This seems rather improbable if not the gene order in all the species has been constant during the evolution of the Leguminosae. It seems therefore more plausible that the multiple effects in the related cases of unifoliate mutants are caused by true pleiotropic gene action.

SIRKS (1931) used three lines of white-flowered plants in his exhaustive examination of the inheritance of different characters in *Vicia faba*. These lines were found to be multiple alleles at one locus. When crossing them to lines with black spots on the wings (designated as normal or standard type in the present investigation), SIRKS obtained a clear monofactorial inheritance. The same results were obtained by DARLINGTON (1929), ERITH (1930), ROWLANDS et al. (1962), and PICARD (1963) in similar investigations. ROWLANDS used a variety, Triple White, which is equal to allele *sp-b*^{1} in the present investigation, whereas the origin of the lines in the other investigations cannot be verified. The important fact in this connection is, however, the corroborative results that were obtained, which indicate that the deviations from the 3 : 1 ratio found in the present analysis of the *sp-b* locus are most likely due to some error in the experiment. The reason for this is, as pointed out earlier, not known, but this analysis will be repeated. Another still more deviating result was reported by GOYAL (1965). He found unpigmented flowers to be dominant over pigmented ones.

In addition, SIRKS (1931) found that the factor reponsible for the expression of black spots on the wings also caused the black spots on the stipules. In recessive condition, this allele also was the one evoking white hilum in the seed and absence of pigments in the seed-coat. The last-mentioned character implies that the seeds, after storage, retain the cinereous colour in contrast to normal seeds, in which the buff colour becomes brown after storage. These effects of the *sp*-gene have also been observed in the present material.

As described earlier, the intensity of the colours in the standard lines is variable. SIRKS (1931) examined the variation of the violet lines and found that it was genetically determined. Three factors were isolated, two of which were multiple alleles affecting the reddish colours and one affecting the bluish colours of the lines. These factors could only be expressed in the presence of the basic factor in the dominant state.

Finally, it should be mentioned in connection with the inheritance of flower colour in *Vicia faba* that ROWLANDS et al. (1962) found a 9 : 3 : 4 ratio in the F_2, i.e., the same ratio that BOND (pers. comm.) and I have observed, when they crossed three coloured types to white-flowered types (*dp-a*×*sp-a*) – according to the symbols used in the present investigation. The other two mutants (identical with *dp-a*^{27} and *dp-a*^{28}) segregated in deviating frequencies, which was interpreted as a 15 : 1 segregation (ROWLANDS et al. l.c.). They obtained this ratio by considering the normal flower as coloured and then adding all coloured types.

The inheritance of flower colour has also been investigated in *Vicia villosa* and *Vicia sativa* (DONNELLY 1958; CLARK and DONNELLY 1964). The normal purple flower colour of both species was found to be dominant over white flowers. A pink-flowered mutant in *Vicia sativa* was crossed to the white-flowered type and gave a purple-flowered F_1 generation and a segregation in the F_2 generation of 9 purple : 3 pink : 4 white flowers. The inheritance of these characters is thus in agreement with the inheritance of corresponding characters in *Vicia faba*. Comparable genetic analyses have been performed in a number of other species in the Leguminosae. It would be of interest to compare the data from all such studies, but this is outside the scope of the present paper.

Besides the paper by SIRKS (1931), the available information from genetic analyses in *Vicia faba* is limited. Earlier, some papers have been discussed in which data on flower colour, seed-coat colour, and hilum colour – the best investigated characters—were presented. Colourless and black are the commonest hilum colours, but TAVCAR (1963) also had isolated plants with a brown hilum, which he found to be conditioned by a multiple allele to black hilum, and thus located in the third linkage group. The brown hilum colour was recessive to black and dominant to colourless, or cinereous.

GOYAL (1965) studied the breeding behaviour of anthocyanin pigmentation in the flower and stem. For both characters, the unpigmented allele was dominant over the pigmented one. Both results are deviating, as has been discussed for flower colour, and in comparison with the present data in which red-stemmed seedling (*Rs*) was

dominant over the normal green-stemmed seedling allele. GOYAL (1966) has furthermore investigated four other characters, which, however, are very briefly presented. He observed that a high number (5) of flower buds in an axil was dominant to a low number (3); that black fruit colour was dominant over green fruit colour with large black spots; that resistance to aphid attacks was dominant over susceptibility; and that resistance to mildew was also dominant to susceptibility.

The practical significance of the induced mutations

Early mutants are one of the most common types among positive mutations, e.g., mutations that are of practical significance. Such mutants are observed in many agricultural crops, for example, barley (GUSTAFSSON et al. 1960), rice (TANAKA 1969), soybean (ZACHARIAS 1966), peas (GOTTSCHALK 1960; WELLENSIEK 1961), lupines (TEDIN 1954; PORSCHE 1966), and oats (MATTSON pers. comm.). In some cases, early mutants have become of direct practical importance, and 11 such released varieties, including wheat, barley, soybean, navy bean, string bean, and castor bean, are described by SIGURBJÖRNSSON et al. (1969).

Very early mutants (about 1 week or more earlier than the parental variety) are often characterized by a reduced stature and with concomitant changes and affects on node number, leaf area, dry weight, thousand-grain weight, and thus in many cases yield. Thus, DUDITS et al. (1970) discussed a mutant in peas that was 9 days earlier and had a height of 42 cm and 14 nodes as compared with a height of 61 cm and 21 nodes in the parental variety. ZACHARIAS (1966) indicated the "low habit" of early types in soybean, and TANAKA (1969) stated that in rice "mutants with extremely early maturity had shorter culms and panicles". The presently available early mutant in *Vicia faba* was found to be significantly shorter than the parental variety and also to have a significantly lower number of nodes. GUSTAFSSON et al. (1960) reported that out of eight extremely early mutants that were investigated in barley, none had a longer straw and three of them had definitely shorter culms than the parental variety. Seven of the mutants had a reduced number of nodes; and when five of the mutants were tested in yield trials, all were inferior to the parental variety. This was under the growing conditions in southern Sweden. However, when one of the early mutants, which later was released as Mari, was tested in central Sweden, it competed well with the parental variety. This difference in sensitivity to day-length conditions has been verified in phytotron experiments (DORMLING et al. 1966), thus indicating that Mari is inferior under conditions that are optimal for the parental variety. But the reactions are changed in favour of Mari when the two varieties are grown under other conditions.

That the optimum conditions are not necessarily the same for the early mutant and its parental variety is discussed further for the early mutant 46 in peas (GOTTSCHALK 1960). This mutant was inferior to the parental variety under wet conditions, but had a higher yield under dry conditions.

The tendency towards lower plant stature probably indicates that early mutants also require different growing conditions than those of the parental varieties to reach an optimal level of yield. At least this is the case for the early mutant in *Vicia faba*, Sv 0420. When grown under the same conditions as Primus, the parental variety, Sv 0420 gives about 10 per cent lower yield per ha, whereas this difference was lowered when the plant density was increased. The hitherto performed experiments show a levelling in the yield relations between the parental variety and the mutant when the seed rate was increased from 200 kg per ha. (the normal seed rate for Primus) to 250 kg per ha. It seems likely that the yield difference could be equalized with a still higher seed rate, since the data indicates that 250 kg per ha. was not giving the optimum plant density for the early mutant.

All of these examples emphasize the importance of considering early mutants as specific types with regard to ecological conditions. It is therefore necessary to grow the early mutant under conditions that are as optimal as possible for the mutant before making any appraisal of its usefulness. If a comparison of the mutant and the parental variety is made under the growing conditions that are optimum for the parental variety, then, the risk of underestimating the value of the mutant is naturally great. This is also likely the case for several other types of mutants as well.

Investigations of induced protein mutations, comprising changes in protein content as well as quality, have increased considerably during recent years. The discovery of the high-lysine mutants, *opaque-2* and *floury-2*, in maize (MERTZ et al. 1964; NELSON et al. 1965), as well as the development of new screening techniques, e.g., the DBC-method (MOSSBERG 1969) provided new impetus. Changes in protein content have been found in mutant collections that were isolated for other characters, e.g., in rice mutants (TANAKA 1969) and in the extensive collection of pea mutants (GOTTSCHALK et al. 1970). In such self-fertilizing species, the identification of the changes in protein content or quality as mutations is clear-cut, while on the other hand, difficulties are met with in determining whether genes with pleiotropic action have mutated or if a group of genes have mutated simultaneously. In a cross-fertilizing species like *Vicia faba* in which a population shows a variation of about 10 per cent units in protein content, it will be rather speculative to state or conclude that a high protein content in a plant treated with mutagens is a result of mutation or an expression of favourable recombinations of existing genes in the parental population. However, such a determination is only of theoretical interest, for from the plant breeding point of view what is most important is the possibility to be able to raise a population with increased protein content. This fact makes, e.g., Sv 0720, described earlier, of interest whatever the mode of origin. Moreover, it is of interest to note that improvement in protein content can be achieved in mutants that have undergone sharp changes in other characters which in turn provide a stimulus for further investigations of mutant collections.

A negative correlation between thousand-kernel weight and protein content has been observed in rice and barley (TANAKA 1969; MUNCK et al. 1970). This was not the case in *Vicia faba*, as pointed out earlier, probably depending upon the differences in seed morphology.

The expression "positive mutation" is mostly referred to mutant types exhibiting an improvement in some characters of direct agricultural value, e.g., earliness, straw stiffness, improved protein content or quality, resistance to diseases, etc. But, it might be possible that certain mutants in the mutation collection, which are only considered as of theoretical interest, in some specific connection might be of importance for the plant breeder. It is therefore urgent for him to work in close co-operation with workers dealing with mutations or preferably that the plant breeder himself handle the mutation collection in order to become familiar with the entire spectrum of variation in the material. An example of such a mutant is the type with a terminal inflorescence in *Vicia faba*. At the time of isolation, this type was considered to be an interesting curiosity, but later on it was found to be of practical significance – as has been stated earlier.

In this regard, it should be remembered that good gene markers also might be of importance from a plant breeding point of view. Thus, the flower colour mutants in *Vicia faba* have been extensively used in investigations of outcrossing frequencies under field conditions and also for estimating minimum distances between populations (to eliminate outcrossing) for seed multiplication.

These investigations have shown an outcrossing frequency of 30 to 35 per cent under field conditions. From the practical standpoint the minimum distance between plots seems to be 100 metres. Beyond this distance, there is, however, up to 0.5 per cent outcrossing – even up to rather large distances, probably the result of pollen transfer by a bee, which happens to reach a distant population (SJÖDIN, unpubl.).

The examples of positive mutations discussed thus illustrate the usefulness of induced mutations as one way of creating new variation in the breeding material. This concerns characters that do not apparently occur in nature (e.g., the type with a terminal inflorescence), as well as characters that already exist (e.g., early mutants). The advantage of the induced early mutants is that the character is directly introduced into an agriculturally valuable type. By introduction of foreign varieties, it is possible to obtain a more extreme earliness but when crossing them to good agricultural types under our conditions, undesired characters have also been transferred. These have especially been found to influence yield in a negative direction.

Moreover, it is evident that even if Sv 0420 shows such positive characters that it will be released as a new agricultural variety, the most important use of the available mutants of *Vicia faba* will be as parents in conventional cross-breeding programs. The data that have been obtained up to now from the mutation studies are

also an incentive to continue the search for new mutation types that potentially could be useful in the breeding program of this agricultural crop.

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